Academic Chinese Vocabulary Testing System

Based on Large Language Models

Design Logic and Preliminary Practices

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Context: Growing Need for Academic Chinese Testing

- Rising international student numbers and younger learners drive demand for academic Chinese skills.
- Learners must read/write academic texts and master specialized vocabulary.
- HSK 7-9 addresses this need, helping universities assess readiness for degree programs.

Vocabulary in academic contexts also plays a crucial role.

Research Background: Academic Chinese Vocabulary Testing

- Academic Chinese testing research started relatively late, currently focused mainly on word list development.
- Current Status:

- Domain-specific word lists developed (e.g., medicine, economics, law).
- Few general academic word lists exist (Zhu, 2020; Li, 2024).

Standardized assessment tools are urgently needed.

Research Purpose

- **Objective**: Drawing on the Chinese Academic Word List (Li, 2024), this study endeavors to develop a **Chinese Academic Vocabulary Size Test**, which offers a reliable reference for evaluating learners' academic Chinese proficiency.

- **Innovation**:
  - Pioneering the integration of **Large Language Models (Deepseek)** into academic Chinese vocabulary assessment
  - Designing an **AI-assisted automated item generation framework**: contextualized generation + collocational constraints + intelligent distractor filtering

- **Significance**: Addressing the lack of academic Chinese testing tools, and validating the **effectiveness and grading reliability** of the system

Research Questions

1. How can LLMs like DeepSeek generate test items based on the Chinese Academic Word List?

2. What quality controls are needed for AI-generated items (context, distractors, difficulty)?

3. How effective is this system as a standardized academic vocabulary assessment tool?

Research Process

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Step 1: Vocabulary extraction

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Step 2: AI-assisted initial item generation

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Step 3: Expert Review & Iterative Adjustment

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Step 4: Validation with criterion measures(planned)

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Step 5: Large-scale testing with Rasch modeling(planned)

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Step 6: System Development & Adaptive Testing(planned)

Step 1 – Vocabulary extraction

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- **Extraction Rules:**
  - Words selected by part of speech proportion (noun/verb/adjective)
  - Focus on bi-syllabic words.

Target Word	Part of Speech	Number of Syllables	Left Collocation	Left Collocation Frequency	Right Collocation	Right Collocation Frequency	Academic Corpus Frequency	Definition
采用	动	2	本文 等 研究 可以 可 中 通常 方法 ...	176 130 90 82 76 70 67 63 ...	了 不同 和 两 单 软件 多 法	164 59 33 31 30 29 28 27 ...	0.0599	【采用】cǎiyòng 动认为合适而使用：～新工艺｜～举手表决方式｜那篇稿子已被编辑部～。

Step2: AI-assisted initial item generation

- **Contextualization:**
  - Collocational constraints from word list
  - Short academic sentences (5–9 characters)
- **Item Format:**
  - 4-option multiple-choice, choose synonym closest in meaning

- **Contextualization**

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我想测试汉语学习者的学术汉语水平（参考hsk7-9级要求及样卷），请用目标词“分类（动词）”造3个5-9个字以内的句子，可以是短语（比如可以缺少主语等）。该目标词在学术语境中常见的右搭配为：“方法”。要求：用上其右搭配“方法”，学术语境或者句子偏学术语体，所造的句子不能使用特定专业领域内的术语，需要为通用学术语境。

Task:

Test academic Chinese proficiency aligned with HSK 7-9 by creating three language items using the target word “分类” (classify).

Criteria:

1. Format: Phrases or sentences (subjects optional).
2. Length: 5-9 Chinese characters per item.
3. Collocation: Must include the right collocate “方法” (method).
4. Style: Formal academic tone.
5. Vocabulary: General academic lexicon; avoid domain-specific terms.

- **Item Format: Design the correct option**

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我想测试汉语学习者的学术汉语水平（参考hsk7-9级要求及样卷），请问在句子“新的分类方法”中，与“分类”近义的词有哪些？需要满足：1、双音节词；2、与目标词学术义相近；3、与目标词词性及用法一致；4、与目标词的语义色彩一致；5、在所给的句子中可以完美替换。

Task:
Identify near-synonyms for “分类(to classify). ” in the phrase “新的分类方法(the new classicization methods)" that meet all criteria below:

Criteria:
1. Disyllabic
2. Academic meaning match
3. Consistent part of speech & usage (verb, modifies "方法")
4. Matching semantic connotation (neutral/formal)
5. Perfect substitutability in the original phrase

- **Item Format: Design the distractors**

Task:

Create 3 distractors for the target word "适用" (applicable) in the sentence "该方法同样适用(This method is also applicable.)," where the correct synonym is "可行" (feasible).

Criteria:

1. Disyllabic words.
2. Part of speech consistent with the target word and its synonym.
3. Semantic connotation aligned with the target word and its synonym.
4. Contextually appropriate in the designed sentence.

Example:

适用：该方法同样适用。

A. 高效

B. 可行（Correct Answer）

C. 有益

D. 恰当

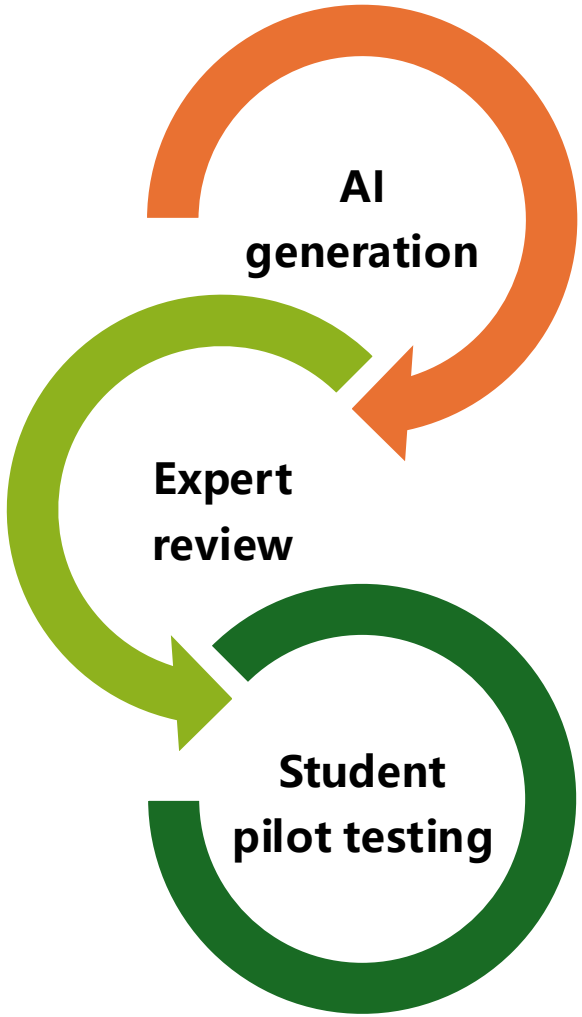
Step 3 – Expert Review & Iterative Adjustment

- Experts evaluate items for:
  - Academic relevance
  - Semantic accuracy
  - Clarity and difficulty
- Items iteratively refined based on expert feedback

Step 4 – Validation (still in progress)

- Small-scale trials with students
- Correlation with:
  - Academic Chinese reading test (self-designed) [Under trial]
  - HSK level
- Purpose: assess practical utility and reliability of test system

Small-scale trials



The Pilot Testing Process

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First Pilot Testing

- Number of items: 210
- Participants: 6
- Native Languages: Vietnamese, Malay, Indonesian, Korean, Japanese, Czech
- HSK Level: All at HSK Level 6

Second Pilot Testing

- Number of items: 185
- Participants: 6
- Native Languages: Vietnamese, Malay, Indonesian, Korean, Japanese, Czech
- HSK Level: 4 at **HSK Level 6**; 2 at **HSK Levels 7–9**

Third Pilot Testing

Number of items: 150
Participants: 3
Native Languages: Vietnamese, Malay
HSK Level: All at **HSK Levels 7–9**

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Step 5: Large-scale testing with Rasch modeling (planned)
Step 6: System Development & Adaptive Testing

Large-scale Administration → Rasch Analysis → System Development → Adaptive Testing+AI

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